

Chapter 10 Practice Test NC#1-9 (20 min) CA #10-16 (20 min)

1. Find $\frac{dy}{dx}$ for $x = \sin \theta$, $y = \tan \theta$ at $\theta = \frac{\pi}{3}$.

a) $\frac{1}{8}$

b) $\frac{1}{2}$

c) $\frac{2}{\sqrt{3}}$

d) 2

e) 8

2. Find the equation of the tangent line for the curve given by $x = 2t$ and $y = t^2 + 5$ at the point where $t = 1$.

a) $y = \frac{7}{2}x - 1$

b) $y = x + 4$

c) $y = -x + 8$

d) $y = 3x$

e) $y = \frac{1}{2}x + 5$

3. What is the arc length of the curve $x = \cos(2t)$ and $y = \sin(2t)$ from $0 \leq t \leq \frac{\pi}{2}$?

a) 0

b) $\frac{\pi}{2}$

c) 2

d) π

e) 2π

4. A particle moves on a plane curve so that at any time t such that $t > 0$, its x -coordinate is $t^3 - t$ and its y -coordinate is $(2t - 1)^3$. Find the acceleration vector of the particle at $t = 1$.

a) $\langle 2, 3 \rangle$

b) $\langle 6, 3 \rangle$

c) $\langle 6, 6 \rangle$

d) $\langle 6, 12 \rangle$

e) $\langle 6, 24 \rangle$

5. If $x = t^2$ and $y = \ln(t^2 + 1)$, then at $t = 1$, what is the value of $\frac{d^2y}{dx^2}$?

a) $-\frac{1}{4}$

b) $-\frac{1}{2}$

c) $\frac{1}{4}$

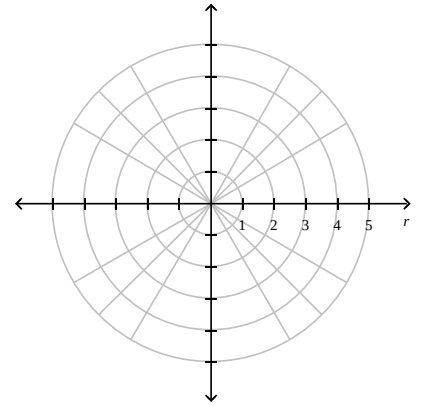
d) $\frac{1}{2}$

e) 2

6. For what value(s) of t , $t > 0$ does the curve given by the parametric equations $x = t^3 - 4t^2 + 5t + 2$ and $y = t^4 - 8t + 5$ have a vertical tangent?

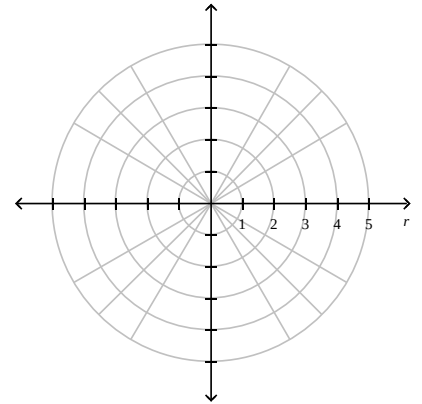
- a) $\sqrt[3]{2}$ b) $\sqrt{2}$ c) 1 d) $\frac{5}{3}$ e) 1 and $\frac{5}{3}$

7. What is the area of the closed region bounded by the polar graph of $r = \sqrt{1 + \cos \theta}$?



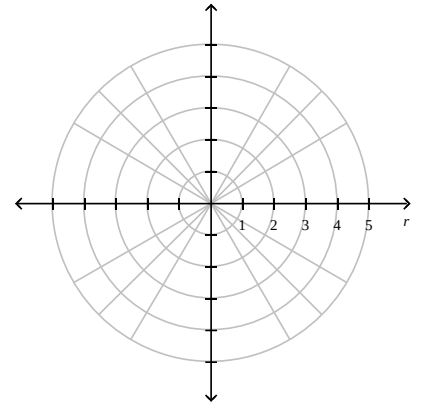
- a) $\int_0^{2\pi} \sqrt{1 + \cos \theta} d\theta$ b) $\int_0^{\pi} \sqrt{1 + \cos \theta} d\theta$ c) $2 \int_0^{2\pi} 1 + \cos \theta d\theta$ d) $\int_0^{\pi} 1 + \cos \theta d\theta$ e) $2 \int_0^{\pi} \sqrt{1 + \cos \theta} d\theta$

8. Which describes the area of the region inside the polar curve $r = 4 \sin \theta$ and outside $r = 2$?



- a) $16 \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sin^2 \theta d\theta + \int_0^{\frac{\pi}{6}} 4 d\theta$ b) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 4 d\theta + 16 \int_0^{\frac{\pi}{6}} \sin^2 \theta d\theta$ c) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 2 d\theta + 8 \int_0^{\frac{\pi}{6}} \sin^2 \theta d\theta$
- d) $\frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (16 \sin^2 \theta - 4) d\theta$ e) $\frac{1}{2} \int_0^{\pi} (16 \sin^2 \theta - 4) d\theta$

9. Which describes the area of the region in common between the polar curve $r = 4\sin\theta$ and $r = 2$?



- a) $16\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sin^2 \theta d\theta + \int_0^{\frac{\pi}{6}} 4d\theta$ b) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 4d\theta + 16\int_0^{\frac{\pi}{6}} \sin^2 \theta d\theta$ c) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 2d\theta + 8\int_0^{\frac{\pi}{6}} \sin^2 \theta d\theta$
- d) $\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (16\sin^2 \theta - 4)d\theta$ e) $\frac{1}{2}\int_0^{\pi} (16\sin^2 \theta - 4)d\theta$

10. Find the equation of the tangent line to the graph of $r = 2\sin\theta$ at $\theta = \frac{\pi}{6}$.

- a) $y - \frac{1}{2} = \frac{1}{\sqrt{3}}\left(x - \frac{\sqrt{3}}{2}\right)$ b) $y - \frac{\sqrt{3}}{2} = \frac{1}{\sqrt{3}}\left(x - \frac{1}{2}\right)$ c) $y - \frac{\sqrt{3}}{2} = \sqrt{3}\left(x - \frac{1}{2}\right)$
- d) $y - 1 = \sqrt{3}\left(x - \frac{\pi}{6}\right)$ e) $y - \frac{1}{2} = \sqrt{3}\left(x - \frac{\sqrt{3}}{2}\right)$

11. What is the length of the polar curve $r = 2 + 2\cos\theta$ from $0 \leq \theta \leq \pi$?

- a) 5.429 b) 6.283 c) 7.271 d) 8 e) 8.886

12. What is the length of the curve determined by $x = 3t$ and $y = 2t^2$ from $0 \leq t \leq 9$?

- a) 504.140 b) 22.906 c) 65.725 d) 20.911 e) 166.139

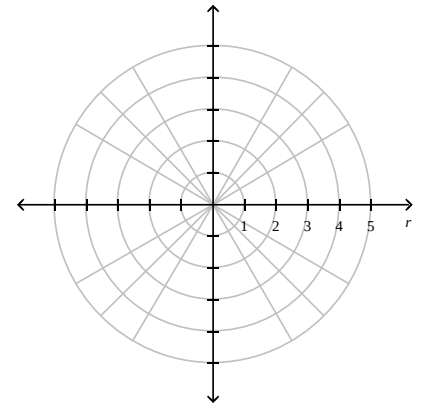
13. If a particle moves in the xy -plane so that at time $t > 0$ its velocity vector is $\langle t^2 \sin(3t - 2), \ln|2^t + 1| \rangle$, then at time $t = \frac{\pi}{2}$, the acceleration vector is...

- a) $\langle -5.423, 0.519 \rangle$ b) $\langle -2.423, 1.519 \rangle$ c) $\langle 0.857, 1.592 \rangle$ d) $\langle 3.857, 2.592 \rangle$ e) $\langle 6, 2 \rangle$

14. If a particle moves in the xy -plane so that at time $t \geq 0$ its velocity vector is $\langle t^2 \sin(3t - 2), \ln|2^t + 1| \rangle$, then what is the position vector at time $t = \frac{\pi}{2}$, given the particle is at $\langle 3, 1 \rangle$ at $t = 0$.

- a) $\langle -5.423, 0.519 \rangle$ b) $\langle -2.423, 1.519 \rangle$ c) $\langle 0.857, 1.592 \rangle$ d) $\langle 3.857, 2.592 \rangle$ e) $\langle 6, 2 \rangle$

15. The area of one loop of the graph of the polar equation $r = 2 \sin(3\theta)$ is given by which of the following expressions?



- a) $4 \int_0^{\frac{\pi}{3}} \sin^2(3\theta) d\theta$ b) $2 \int_0^{\frac{\pi}{3}} \sin(3\theta) d\theta$ c) $2 \int_0^{\frac{\pi}{3}} \sin^2(3\theta) d\theta$ d) $2 \int_0^{\frac{2\pi}{3}} \sin^2(3\theta) d\theta$ e) $2 \int_0^{\frac{2\pi}{3}} \sin(3\theta) d\theta$

16. What is the total distance travelled by a particle moving along the parametric equation $x = \arctan(2t + 1)$, $y = 3 \ln|4t + 1|$ from $1 \leq t \leq 3$?

- a) 13.258 b) 2.872 c) 2.868 d) 0.722 e) 0.739

CA 1. An object moving along a curve in the xy -plane is at position $\langle x(t), y(t) \rangle$ at time t , where

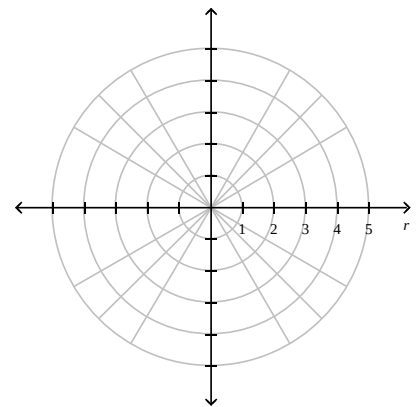
$$\frac{dx}{dt} = \sin^{-1}(1 - 2e^{-t}) \quad \text{and} \quad \frac{dy}{dt} = \frac{4t}{1+t^3}$$

for $t \geq 0$. At time $t = 2$, the object is at the point $(6, -3)$.

- Write an equation for the line tangent to the curve at $(6, -3)$.
- Find the speed of the object at time $t = 2$ and the distance travelled from $0 \leq t \leq 2$.
- Is the object moving closer or farther away from the x -axis at $t = 2$? Explain your reasoning.
- Find the position of the object at time $t = 3$.

NC 2. Consider the polar curve $r = 2 - 2\sin\theta$. Note: $\sin^2\theta = \frac{1 - \cos 2\theta}{2}$

- Find the area of the region enclosed by the curve.
- Write the equation of the tangent line to the curve at $\theta = \frac{\pi}{6}$.
- Setup, but don't evaluate, an expression representing the length of the curve from $0 \leq \theta \leq \frac{\pi}{2}$.



Answers

1. $\frac{dy}{dx} = \frac{y'(\theta)}{x'(\theta)} = \frac{\sec^2 \theta}{\cos \theta}$. Evaluating this at $\theta = \frac{\pi}{3}$ gives us $\frac{4}{0.5} = 8$ so **E**.

2. The equation of the tangent line needs a slope and a point. The point at $t = 1$ is $(x(1), y(1)) = (2, 6)$ and the slope is $\frac{dy}{dx} = \frac{y'(1)}{x'(1)} = \frac{2t}{2} = t|_{t=1} = 1$. So the equation is $y - 6 = 1(x - 2)$ or **B**.

3. Arc length is $\int_0^{\frac{\pi}{2}} \sqrt{\left(\frac{dy}{dt}\right)^2 + \left(\frac{dx}{dt}\right)^2} dt$, so $\frac{dy}{dt} = 2\cos 2t$ and $\frac{dx}{dt} = -2\sin 2t$. So,

$$\int_0^{\frac{\pi}{2}} \sqrt{(2\cos 2t)^2 + (-2\sin 2t)^2} dt = \int_0^{\frac{\pi}{2}} \sqrt{4\cos^2 2t + 4\sin^2 2t} dt = \int_0^{\frac{\pi}{2}} \sqrt{4(\cos^2 2t + \sin^2 2t)} dt.$$

This has a Pythagorean identity, which can be rewritten as $\int_0^{\frac{\pi}{2}} \sqrt{4(1)} dt = \int_0^{\frac{\pi}{2}} 2 dt$. Integrating gives us $2t|_0^{\frac{\pi}{2}} = \pi$ or **D**.

4. Given the position, we must differentiate twice to get acceleration. The position vector would be $\langle t^3 - t, (2t - 1)^3 \rangle$ and the velocity vector would be $\langle 3t^2 - 1, 6(2t - 1)^2 \rangle$ and the acceleration vector would be $\langle 6t, 24(2t - 1) \rangle$. At $t = 1$, the acceleration vector would be $\langle 6, 24 \rangle$ or **E**.

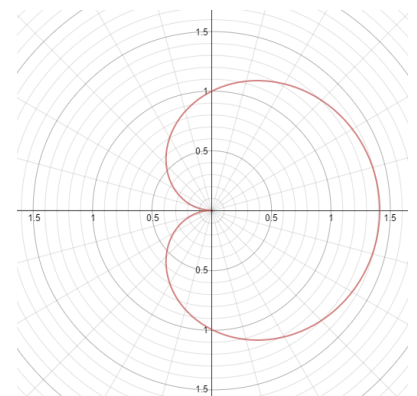
5. $\frac{d^2 y}{dx^2} = \frac{\frac{d}{dt} \left[\frac{dy}{dx} \right]}{\frac{dx}{dt}}$ and $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$. Therefore, $\frac{dy}{dt} = \frac{2t}{t^2 + 1}$ and $\frac{dx}{dt} = 2t$, so $\frac{dy}{dx} = \frac{\frac{2t}{t^2 + 1}}{2t} = \frac{1}{t^2 + 1}$. The second

derivative would then be $\frac{d^2 y}{dx^2} = \frac{\frac{d}{dt} \left[\frac{1}{t^2 + 1} \right]}{\frac{dx}{dt}} = \frac{\frac{-2t}{(t^2 + 1)^2}}{2t} = \frac{-1}{(t^2 + 1)^2} \bigg|_{t=1} = -\frac{1}{4}$ or **A**.

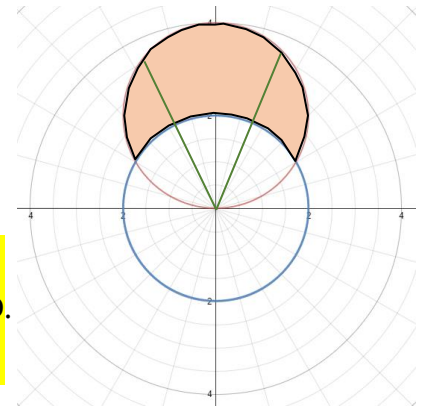
6. Vertical tangent lines occur when $\frac{dx}{dt} = 0$. So, $\frac{dx}{dt} = 3t^2 - 8t + 5 = 0$ or $(3t - 5)(t - 1) = 0$ or $t = \frac{5}{3}, 1$ or **E**.

7. A quick sketch is shown to the right. This graph goes from $[0, 2\pi]$ with symmetry from $[0, \pi]$. So the area is $\frac{1}{2} \int_a^b r^2 d\theta = 2 \cdot \frac{1}{2} \int_0^{\pi} (\sqrt{1 + \cos \theta})^2 d\theta$

or $\frac{1}{2} \int_a^b r^2 d\theta = \frac{1}{2} \int_0^{2\pi} (\sqrt{1 + \cos \theta})^2 d\theta$. This gives us $\int_0^{\pi} (1 + \cos \theta) d\theta$ or **D**.

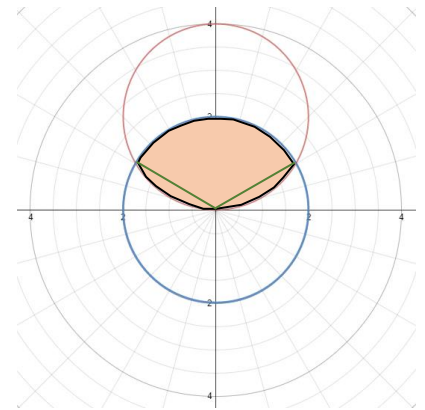


8. A quick sketch is shown to the right. This region goes from the intersection of the two curves. This is when $2 = 4\sin\theta$, $\sin\theta = \frac{1}{2}$, $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ with symmetry from $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$. As shown by the green lines, $r = 4\sin\theta$ is always



greater and $r = 2$ is always smaller, so the area is $\frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} [(4\sin\theta)^2 - (2)^2] d\theta$ or **D**.

9. A quick sketch is shown to the right. This region goes from $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$ or $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$ with symmetry. As shown by the green lines, this region switches which curve is on the outside (there is no inside function). From $\left[0, \frac{\pi}{6}\right]$, $r = 4\sin\theta$ is on the outside and then from $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$, $r = 2$ is on the outside.



Therefore, we need 2 integrals, so $2 \cdot \left[\frac{1}{2} \int_0^{\frac{\pi}{6}} (4\sin\theta)^2 d\theta + \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (2)^2 d\theta \right]$ or **B**.

10. We need slope and a point to write the equation of a tangent line. Using $x = r \cos\theta$, $y = r \sin\theta$ gives us the point $x = 2\sin\left(\frac{\pi}{6}\right)\cos\left(\frac{\pi}{6}\right)$, $y = 2\sin\left(\frac{\pi}{6}\right)\sin\left(\frac{\pi}{6}\right)$ or $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$. The slope is $\frac{dy}{dx} = \frac{4\sin\theta\cos\theta}{2\cos^2\theta - 2\sin^2\theta}$ or at $\theta = \frac{\pi}{6}$ is $\frac{dy}{dx} = \frac{\sqrt{3}}{1}$. Therefore the equation is $y - \frac{1}{2} = \sqrt{3}\left(x - \frac{\sqrt{3}}{2}\right)$ or **E**.

11. Arc length is $\int_a^b \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$, so we need the derivatives of $x = r \cos\theta$, $y = r \sin\theta$. So,

$$x = (2 + 2\cos\theta)\cos\theta = 2\cos\theta + 2\cos^2\theta, \quad y = (2 + 2\cos\theta)\sin\theta = 2\sin\theta + 2\sin\theta\cos\theta \quad \text{and}$$

$\frac{dx}{d\theta} = -2\sin\theta - 4\cos\theta\sin\theta$, $\frac{dy}{d\theta} = 2\cos\theta - 2\sin^2\theta + 2\cos^2\theta$. Plugging $\int_0^{\pi} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$ into the calculator (I recommend using y_1, y_2) gives us 8 or **D**.

12. Arc length is $\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$, so we need the derivatives of $x = 3t$ and $y = 2t^2$, which are

$$\frac{dx}{dt} = 3, \frac{dy}{dt} = 4t. \text{ So, using the calculator, } \int_0^9 \sqrt{(3)^2 + (4t)^2} dt = 166.139 \text{ or E.}$$

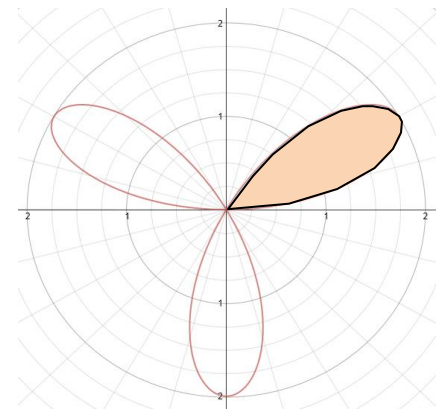
13. To find acceleration given velocity, we must differentiate. Since this is a derivative at a point, we can do it on the calculator and just do each component individually. This gives us $\langle -5.423, 0.519 \rangle$ or A.

14. To find position given velocity, we must integrate. Since we have two bounds, we can do it on the calculator and just do each component individually. It is very important to remember the initial condition,

$$\text{so } x\left(\frac{\pi}{2}\right) = 3 + \int_0^{\frac{\pi}{2}} (t^2 \sin(3t - 2)) dt = 3.857 \text{ and } y\left(\frac{\pi}{2}\right) = 1 + \int_0^{\frac{\pi}{2}} (\ln|2^t + 1|) dt = 2.592 \text{ or D.}$$

15. A quick sketch is shown to the right. Each loop begins and ends when the curve passes through $r = 0$. This occurs at $0 = 2 \sin 3\theta$, $\sin 3\theta = 0$ or $3\theta = 0, \pi, 2\pi \dots$ so $\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3} \dots$ Since $r = 2 \sin 3\theta$ is always outside,

$$\text{the area of one loop would be } \frac{1}{2} \int_0^{\frac{\pi}{3}} [(2 \sin 3\theta)^2] d\theta = 2 \int_0^{\frac{\pi}{3}} \sin^2 3\theta d\theta \text{ or C.}$$



16. Distance travelled is the same as arc length, which is $\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$. Finding the derivative of each gives us $\frac{dx}{dt} = \frac{2}{1 + (2t + 1)^2}$, $\frac{dy}{dt} = \frac{12}{4t + 1}$. Plugging $\int_1^3 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$ into the calculator (I recommend using y_1, y_2) gives us 2.872 or B.

FR1:

- a) The equation of a tangent line requires point and slope. We are given the point $(6, -3)$, so we need slope. We are given $t = 2$, so the slope $\left. \frac{dy}{dx} \right|_{t=2} = \frac{y'(2)}{x'(2)} = 1.088$. So, the equation is $y + 3 = 1.088(x - 6)$.
- b) The speed at $t = 2$ is $\sqrt{[x'(2)]^2 + [y'(2)]^2} = 1.208$ and the distance travelled is found using $\int_0^2 \sqrt{[x'(t)]^2 + [y'(t)]^2} dt = 3.200$.
- c) Moving closer or farther away from the x -axis is movement in the y -direction, so we should focus on $\frac{dy}{dt}$. At $t = 2$, the y -coordinate is -3 and $y'(2) = \frac{8}{9}$. Since this is positive, that means the y -coordinates are going up. Therefore, the object is moving closer.
- d) To find the position given the velocity (or derivative), we must integrate and remember the initial condition. Since the x - and y -coordinate are independent of one another, we must do both separately. $x(3) = 6 + \int_2^3 \frac{dx}{dt} dt = 6.983$ and $y(3) = -3 + \int_2^3 \frac{dy}{dt} dt = -2.380$, or $(6.983, -2.380)$.

FR2:

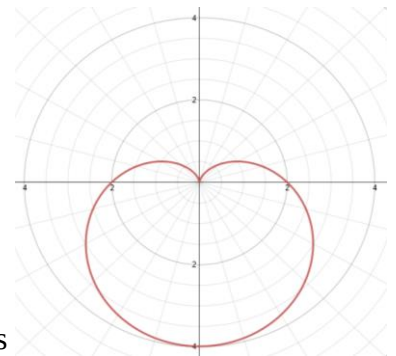
- a) A quick sketch is shown to the right. The curve goes from $[0, 2\pi]$ or uses symmetry from $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$. The area would be $\frac{1}{2} \int_0^{2\pi} (2 - 2\sin\theta)^2 d\theta$.

This can't be integrated as is, so we must multiply it out or

$$\frac{1}{2} \int_0^{2\pi} (4 - 8\sin\theta + 4\sin^2\theta) d\theta, \text{ which using the given trig identity,}$$

$$\frac{1}{2} \int_0^{2\pi} (4 - 8\sin\theta + 2 - 2\cos 2\theta) d\theta = \int_0^{2\pi} (3 - 4\sin\theta - \cos 2\theta) d\theta. \text{ The last needs}$$

$$\text{a u-sub with } u = 2\theta, du = 2d\theta \text{ so } 3\theta + 4\cos\theta - \frac{1}{2}\sin 2\theta \Big|_0^{2\pi} = (6\pi + 4 - 0) - (0 + 4 - 0) = 6\pi.$$



- b) The equation of the tangent line requires a point and slope. Using $x = r \cos \theta$, $y = r \sin \theta$ gives us $x = (2 - 2 \sin \theta) \cos \theta$, $y = (2 - 2 \sin \theta) \sin \theta$. The point would then be

$$x = \left(2 - 2 \sin \left(\frac{\pi}{6}\right)\right) \cos \left(\frac{\pi}{6}\right), y = \left(2 - 2 \sin \left(\frac{\pi}{6}\right)\right) \sin \left(\frac{\pi}{6}\right) \text{ or } \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right). \text{ The slope is}$$

$$\frac{dy/d\theta}{dx/d\theta} = \frac{-2 \cos \theta \sin \theta + (2 - 2 \sin \theta)(\cos \theta)}{-2 \cos^2 \theta + (2 - 2 \sin \theta)(-\sin \theta)} \text{ or at } \theta = \frac{\pi}{6} \text{ is } \frac{dy}{dx} = \frac{-\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}}{-\frac{3}{2} - \frac{1}{2}} = 0. \text{ Therefore, the}$$

equation is $y - \frac{1}{2} = 0 \left(x - \frac{\sqrt{3}}{2} \right).$

- c) Arc length is found with $\int_0^{\frac{\pi}{2}} \sqrt{\left[\frac{dx}{d\theta}\right]^2 + \left[\frac{dy}{d\theta}\right]^2} d\theta$ and using the above derivatives, we get

$$\int_0^{\frac{\pi}{2}} \sqrt{[-2 \cos^2 \theta + (2 - 2 \sin \theta)(-\sin \theta)]^2 + [-2 \cos \theta \sin \theta + (2 - 2 \sin \theta)(\cos \theta)]^2} d\theta.$$